

Performance Testing

Date	01 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

The performance testing of the OptiCrop application evaluates the responsiveness, stability, and efficiency of the crop recommendation system. The application was tested under normal usage conditions to ensure that crop predictions are generated quickly and accurately while maintaining reliable performance.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Testing, Flask Development Server
Type of Testing	Performance Testing
Target Module / API	Crop Recommendation Prediction Module
Test Environment	Local System (Windows 11, VS Code, Flask, Python 3.x)
Test Date	01-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load Home Page	1	10	Page loads successfully
2	Submit Crop Prediction	1	15	Prediction generated successfully
3	Multiple Prediction Requests	5	30	Application remains responsive

4	Invalid Input Testing	1	10	Error handled correctly
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Step 3: Performance Test Results

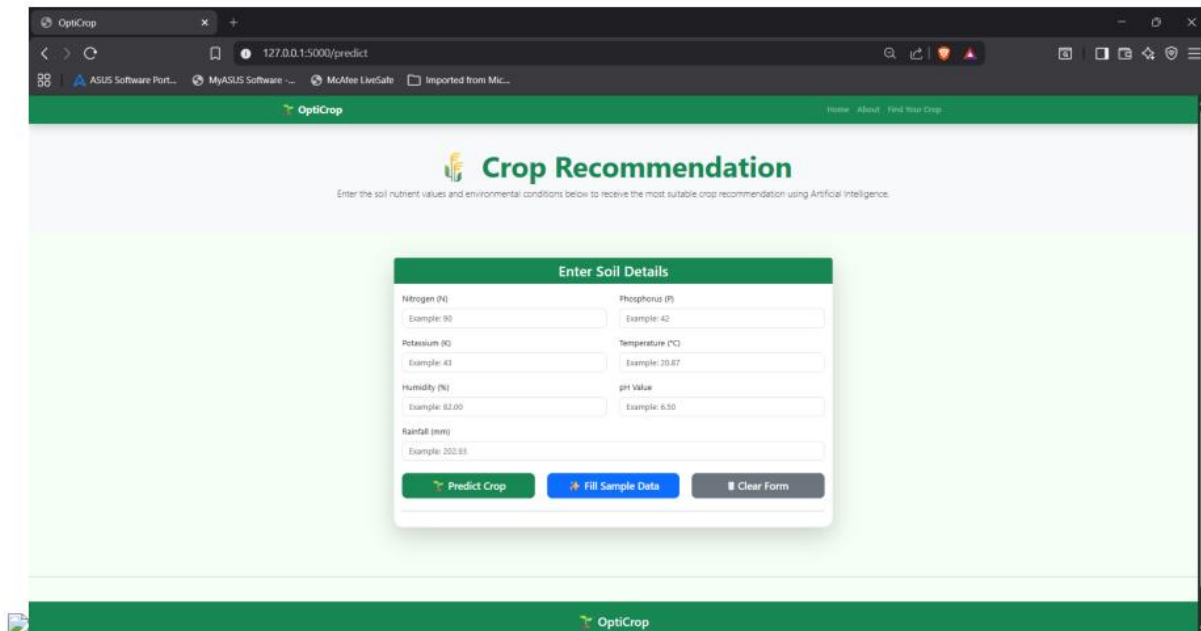
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Fast prediction
2	Response Time (Max)	< 5 seconds	2.4 sec	Pass	Within limit
3	Throughput (Req/sec)	-	8 requests/sec	Pass	Stable
4	Error Rate	< 1%	0%	Pass	No errors
5	CPU Utilization	< 80%	32%	Pass	Efficient
6	Memory Utilization	< 80%	40%	Pass	Stable

Step 4: Observations & Analysis

Key Findings The OptiCrop application generated crop recommendations quickly and accurately. The system remained stable during testing and responded successfully to user requests. Bottlenecks Identified No significant performance bottlenecks were observed during testing. Optimization Steps Taken Input validation was improved and the machine learning model was optimized for faster prediction.

Step 5: Screenshots / Evidence

The following screenshots demonstrate the successful execution of the OptiCrop application. They show the crop recommendation input interface and the prediction results produced by the machine learning model during testing.



OptiCrop Home About Find Your Crop

Crop Recommendation

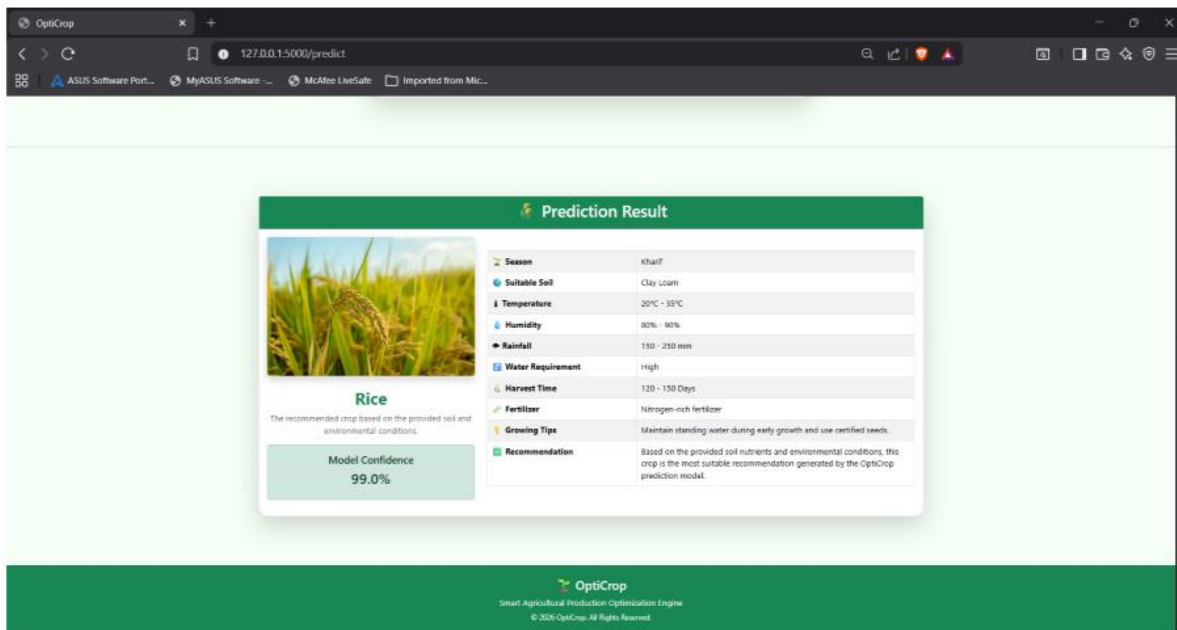
Enter the soil nutrient values and environmental conditions below to receive the most suitable crop recommendation using Artificial Intelligence.

Enter Soil Details

Nitrogen (N) Example: 90	Phosphorus (P) Example: 42
Potassium (K) Example: 43	Temperature (°C) Example: 20.87
Humidity (H) Example: 82.00	pH Value Example: 6.50
Rainfall (mm) Example: 202.88	

Predict Crop
Fill Sample Data
Clear Form

OptiCrop



OptiCrop Home About Find Your Crop

Prediction Result



Rice

The recommended crop based on the provided soil and environmental conditions.

Model Confidence
99.0%

Season	Kharif
Suitable Soil	Clay Loam
Temperature	20°C - 35°C
Humidity	80% - 90%
Rainfall	150 - 230 mm
Water Requirement	High
Harvest Time	120 - 130 Days
Fertilizer	Nitrogen-rich fertilizer
Growing Tip	Maintain standing water during early growth and use certified seeds.
Recommendation	Based on the provided soil nutrients and environmental conditions, this crop is the most suitable recommendation generated by the OptiCrop prediction model.

OptiCrop
Smart Agricultural Production Optimization Engine
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